Cryptography

Lecture 7 – Exam Review

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Outline

1. General Concepts
2. Perfect Encryption
3. Private-Key Crypto
   - Building Blocks
   - Encryption
   - Authentication

Security Definitions

- Game-based security definitions
  - How they capture adversary capabilities
  - What it means for $\mathcal{A}$ to win
  - The use of oracles in the definitions
  - Be able to write a definition given an adversary description

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  - The use of oracles in the definitions
  - Be able to write a definition given an adversary description
- Understand difference between "indistinguishability" and "unforgeability" style definitions
- Relationships between definitions (e.g., CCA is strengthening of CPA)
- Negliguble functions and the birthday paradox

Proofs by Reduction

- Understand proof structure and what it implies
  - Assume existence of adversary $\mathcal{A}_c$ vs. construction
  - Show this implies existence of adversary $\mathcal{A}_r$ vs. assumption
    - Step 1: Build such an $\mathcal{A}_r$
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- Note: Not enough to just draw picture of reduction, have to explain why it works.

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One-Time Pad

- Construction
- Security definition
- Limitations
  - One-time use
  - Key as long as message
  - Be able to argue why these are inherent

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PRGs

- Security definition
  - Length extension
  - Pseudorandomness – as a game
- Be able to argue whether simple constructions are or aren't necessarily PRGs

PRFs

- Security definition
  - Recall what we mean by random function – what is the distribution
  - Indistinguishability from a random function – as a game
  - Oracle notation
- Syntax – distinguish between key and input

Private-Key Encryption

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  - CPA
  - CCA
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- Modes of operations
  - Why we need modes of operations
  - Constructions and key properties (CBC, CTR)
  - Padding oracle attack – why this breaks CCA security

MACs

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- Security definition
- Construction based on PRF

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- Using MACs to build authenticated encryption
  - Encrypt then authenticate

Hash Functions

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- Applications
  - Merkle tree for outsourced storage
  - Hash-and-MAC
  - Passwords
  - Identifiers

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Exam Procedures

- Exam on Mon., Oct. 19, 12:45-3:15 PM in the classroom
- You may bring 2 pieces of 8.5 x 11 inch paper (back and front) with notes
- No computers, phones, or calculators during exam – bring pens or pencils

Exam format

The exam will contain the following:

1. 10 True/False questions – no partial credit
2. 3-4 long answer questions – definitions, reductions, PRG/PRF, Encryption, MACs, Hash functions
3. 1 challenge problem
4. Questions may have multiple parts, complete as much as you can.